COMPLETE PRODUCT & CURRICULUM MASTER BLUEPRINT

JEE Study Website: Step-by-Step Build & NCERT Master Guide

A zero-confusion roadmap to building a JEE Main & Advanced study portal with Class 11 & 12 NCERT chapter mapping, High-Repeat weightage tags, highlighted concept notes, and smart revision tools.

1. Website Architecture & 6-Step Execution Roadmap

Phase 1: Foundation

Ek successful JEE website banane ke liye aapko **Content Structure (NCERT Class 11 & 12)** aur **Clean Tech Stack** ko sahi order mein set karna hoga. Follow this 6-step pipeline to build your platform without technical roadblocks:

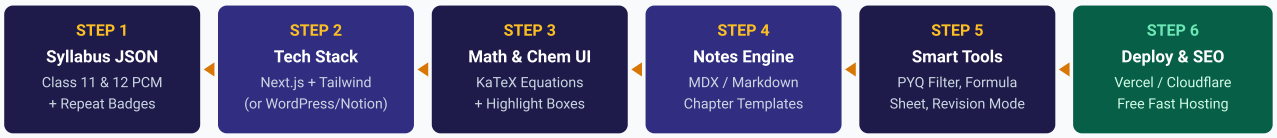


Figure 1.1: End-to-End Development Pipeline for a High-Performance JEE Study Portal

2. Step-by-Step Technical Setup (Choose Your Path)

Implementation

1. **Step 1 — Choose Your Development Path (Coding vs. No-Code):**
   • *Option A (Recommended for Coding/Full Control):* Use **Next.js (React) + Tailwind CSS + MDX**. Ismein har chapter ek simple Markdown (`.mdx`) file hoti hai aur website super-fast load hoti hai. Host for free on **Vercel** or **Cloudflare Pages**.
   • *Option B (Zero-Coding / Fast Launch):* Use **WordPress (with GeneratePress theme + RankMath SEO)** or **Notion + Super.so / Potion**. Agar aap coding mein time waste kiye bina seedha notes daalna chahte hain, toh Notion ya WordPress best hai.
2. **Step 2 — Setup Navigation Hierarchy (NCERT Class 11 & 12 Filter):** Top navbar mein 3 main tabs rakhein: **Physics | Chemistry | Mathematics**. Har subject ke andar 2 toggle buttons dein: `Class 11 NCERT` aur `Class 12 NCERT`, plus ek quick filter button: 🔥 `Show Only High-Repeat Chapters`.
3. **Step 3 — Integrate Scientific Formula Rendering (Crucial for PCM):** Normal text editors mein Physics/Maths ke integration (`∫`), vectors, aur Chemistry ke reversible reactions (`⇌`) toot jaate hain. Integrate **KaTeX** (fastest math renderer) so formulas render crisply on mobile and desktop.
4. **Step 4 — Standardize Chapter URL & Database Structure:** Har chapter ka clean URL rakhein (e.g., `yoursite.com/physics/class-11/rotational-motion`) taaki Google search mein "Rotational Motion JEE short notes" type karne par aapka page top par aaye.

> 💡 **Recommended Folder & Data Schema for Your Website (`syllabus.json`)**
> Apne project mein har chapter ke liye yeh 6 fields zaroor save karein: **1. Subject** (`Physics/Chemistry/Maths`), **2. Class** (`11` or `12`), **3. NCERT Chapter No. & Title**, **4. Repeat Priority Tag** (`🔥 High Repeat [2–3 Qs/Shift]`, `⭐ Medium Repeat [1 Q/Shift]`, or `📘 Foundation`), **5. Highlighted Subtopics List**, and **6. PDF Formula Sheet Link**.

3. Ideal Chapter Note Page Layout (UI/UX Wireframe)

Page Anatomy

Jab koi JEE aspirant kisi chapter par click kare, toh page clutter-free hona chahiye aur **Important / Repeated Topics** turant aankhon ke saamne highlight hone chahiye. Design your chapter page using this proven 3-column academic layout:

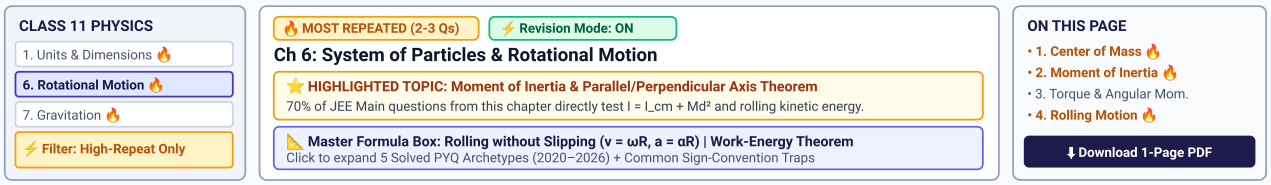


Figure 2.1: 3-Column Distraction-Free Chapter Page Wireframe with Highlighted Subtopics & Quick Jump Links

4. Standard 5-Part Template for Every Chapter's Detailed Notes

Content Standard

- **Part 1: Chapter Weightage & Priority Banner (Top of Page):** Har chapter ke shuru mein likhein ki pichle 5 saalon mein kitne questions aaye hain (e.g., *"Avg 2–3 Questions per paper | Priority: High Repeat 🔥"*) aur kaunse 3 sub-topics sabse zyada repeat hote hain.
- **Part 2: Complete Theory with 4-Color Visual Highlighting:** Plain black-and-white text koi nahi padhta. Use 4 highlight box styles:
  - 🔥 Red/Amber Highlight Box for **Most Repeated JEE Concepts** (jinse bar-bar numericals bante hain).
  - 📘 Green NCERT Line Box for **Direct NCERT Statements & Exceptions** (especially Inorganic, Organic, Biomolecules, Modern Physics).
  - 📐 Indigo Formula Card for **Standard Formulas, Limiting Cases & Units/Dimensions**.
  - ⚠️ Mistake Alert Box for **Common Student Traps** (e.g., forgetting sign convention in thermodynamics work done: Chemistry vs. Physics).
- **Part 3: Reaction / Derivation Mechanism Flowcharts:** Organic Chemistry mein reagent maps aur Physics/Maths mein step-by-step derivation summaries.
- **Part 4: Top 10 Solved Archetype Questions (PYQ Patterns):** Har chapter ke end mein 100 random questions dene ke bajaye, **10 fixed question patterns** dein jo NTA har saal numbers badal kar puchta hai (e.g., Shortest distance between two skew lines in 3D Geometry).

5. Extra High-Value Features ("Apne Se Important Additions")

Secret Sauce

🚀 **6 Game-Changing Features Jo Aapki Website Ko Top Coaching Portals Se Behtar Banayenge**

1. **1-Click "Exam-Eve Revision Mode" Toggle:** Page ke top par ek switch dein. Jab student isko ON kare, toh normal theory chhup jaaye aur **sirf Highlighted Topics, Formulas, aur Exceptions** screen par dikhein! Students exam se 1 hafta pehle is feature ke deewane ho jayenge.
2. **NCERT Line-by-Line Exception Tracker (Chem & Physics):** JEE Main mein Inorganic, Organic aur Semiconductors se seedha NCERT ki lines uthayi jaati hain. Add a dedicated tab called *"NCERT Hidden Gems & Tables"* inside every chapter.
3. **"Mark as Weak / Revise Later" Bookmark (LocalStorage):** Bina login kiye bhi student kisi mushkil topic ko ⭐ `Add to My Weak Topics` mark kar sake (browser ke LocalStorage mein save ho jaaye) taaki woh dashboard par apni weak list dekh sake.
4. **80/20 Pareto High-Yield Tracker:** Ek progress bar dikhayein ki student ne top 35 "Most Repeated Chapters" (jo 75% marks cover karte hain) mein se kitne complete kar liye hain.
5. **Interactive Formula & Reagent Flashcards:** Chapter ke end mein 10 flip-cards dein jisse student turant formulas aur Organic reagents yaad kar sake.
6. **Built-in Dark Mode & Distraction-Free Pomodoro Timer:** Raat ko padhne wale JEE aspirants ke liye soft dark theme (`#0f172a`) aur corner mein 45-min study timer.

6. PHYSICS: Class 11 & 12 NCERT Master Matrix & Highlighted Topics Subject 1: Physics

Apni website ke Physics section mein Class 11 aur Class 12 ke saare NCERT chapters ko neeche diye gaye order mein list karein. Jinke aage 🔥 High Repeat laga hai, unpar website mein "**High Weightage Badge**" lagayein aur unke **Must-Highlight Topics** ko special callout boxes mein rakhein:

Class	NCERT Chapter Name	Repeat Tag & Avg Qs	Must-Highlight Topics in Website Notes (High-Frequency Concepts)
11th	1. Units, Dimensions & Errors	🔥 High (1–2 Qs)	**Dimensional analysis**, Vernier Callipers & Screw Gauge least count, Error propagation.
11th	2. Motion in a Straight Line	⭐ Med (1 Q)	**v–t and a–t graphs**, Motion under gravity, Variable acceleration calculus (v·dv/dx).
11th	3. Motion in a Plane	⭐ Med (1 Q)	**Projectile motion** (range/height equations), River-swimmer & Rain-man relative velocity.
11th	4. Laws of Motion & Friction	🔥 High (1–2 Qs)	Free body diagrams,Pulley-wedge constraints, **Two-block friction**, Circular banking.
11th	5. Work, Energy & Power	🔥 High (1–2 Qs)	**Work-Energy Theorem**, Vertical circular motion, Spring potential energy, Collisions (e).
11th	6. System of Particles & Rotation	🔥 High (2–3 Qs)	**Moment of Inertia (Parallel/Perp axes)**, Cavity COM, Angular momentum conservation, Rolling.
11th	7. Gravitation	🔥 High (1–2 Qs)	Variation of g (height/depth), **Escape & Orbital velocity**, Satellite energy, Kepler's laws.
11th	8. Mechanical Props of Solids & Fluids	⭐ Med (1–2 Qs)	Stress-strain curve, Young's modulus, **Bernoulli's theorem**, Surface tension & Terminal velocity.
11th	9. Thermal Properties & KTG	🔥 High (1–2 Qs)	Calorimetry, Conduction (rod combinations), **RMS/Most probable speeds**, Degrees of freedom (γ).
11th	10. Thermodynamics	🔥 High (2 Qs)	**First Law (ΔQ = ΔU + ΔW)**, Isothermal/Adiabatic/Polytropic work & P–V indicator graphs.
11th	11. Oscillations (SHM) & Waves	⭐ Med (1–2 Qs)	**Spring-block time period**, Energy in SHM, Organ pipes (open/closed harmonics) & Beats.
12th	1. Electric Charges & Fields	🔥 High (1–2 Qs)	Coulomb's vector form, **Electric dipole in field (τ, U)**, Gauss's Law flux applications.
12th	2. Potential & Capacitance	🔥 High (2 Qs)	Equipotential surfaces, **Capacitor networks with dielectrics**, Common potential & heat loss.
12th	3. Current Electricity	🔥 High (2–3 Qs)	**Kirchhoff's Laws & Wheatstone bridge**, Cell combinations, Galvanometer to Ammeter/Voltmeter.
12th	4. Moving Charges & Magnetism	🔥 High (2 Qs)	**Biot-Savart (circular arcs/wires)**, Ampere's Law, Charged particle in B-field (r = mv/qB).
12th	5. Magnetism & Matter + EMI	⭐ Med (1–2 Qs)	Dia/Para/Ferro properties, **Faraday's & Lenz's Law**, Motional EMF (Bvl), Self & Mutual inductance.
12th	6. Alternating Current & EM Waves	🔥 High (2 Qs)	**Series LCR Resonance & Phasors**, Power factor (cos φ), Transformer, **EM Wave E/B ratio**.
12th	7. Ray Optics & Optical Instruments	🔥 High (2–3 Qs)	Mirror/Lens formula, **Total Internal Reflection (TIR) & Prism**, Lens Maker's formula, Microscope/Telescope.
12th	8. Wave Optics	🔥 High (1–2 Qs)	**Young's Double Slit (YDSE) with slab**, Fringe width (β = λD/d), Single-slit diffraction & Malus law.
12th	9. Dual Nature, Atoms & Nuclei	🔥 High (3–4 Qs)	**Photoelectric effect graphs**, de Broglie wavelength, **Bohr model (r, v, E, Rydberg)**, Binding energy.
12th	10. Semiconductor Electronics	🔥 High (1–2 Qs)	p-n junction diode (forward/reverse), Rectifiers, **Zener diode regulator**, **Logic Gates truth tables**.

7. CHEMISTRY: Class 11 & 12 NCERT Master Matrix & Highlighted Topics

Subject 2: Chemistry

Chemistry mein website par 2 filter views dein: **Class-wise (Class 11 vs. Class 12 NCERT)** aur **Branch-wise (Physical, Inorganic, Organic)**. Note that p-Block & Salt Analysis are still in JEE syllabus, so include them with clear NCERT tags:

Class & Type	NCERT Chapter Name	Repeat Tag & Qs	Must-Highlight Topics in Website Notes (High-Frequency Concepts)
11th • Phys	1. Some Basic Concepts (Mole)	🔥 High (1–2 Qs)	**Limiting reagent & Stoichiometry**, Molarity/Molality/Mole fraction interconversion.
11th • Phys	2. Structure of Atom	🔥 High (1–2 Qs)	Bohr model formulas, **Quantum numbers (n, l, m, s)**, Radial/Angular nodes, Electronic config exceptions.
11th • Inorg	3. Classification & Periodicity	⭐ Med (1 Q)	**Ionization enthalpy & Electron gain enthalpy exceptions**, Isoelectronic ionic radii order.
11th • Inorg	4. Chemical Bonding & Structure	🔥 High (2–3 Qs)	**VSEPR Shapes & Hybridization**, Dipole moment, **MOT (Bond order & magnetism)**, H-bonding.
11th • Phys	5. Chemical Thermodynamics	🔥 High (2 Qs)	$\Delta H = \Delta U + \Delta n_g RT$, **Hess's Law & Bond Enthalpy**, Entropy & **Gibbs Free Energy ($\Delta G = \Delta H - T\Delta S$)**.
11th • Phys	6. Equilibrium (Chem & Ionic)	🔥 High (1–2 Qs)	K_p & K_c relation, Le Chatelier's principle, **pH of Buffers, Salt Hydrolysis & Solubility Product (K_{sp})**.
11th • Phys	7. Redox Reactions	⭐ Med (1 Q)	Oxidation number paradoxes (CrO_5, H_2SO_5), **Equivalent weight & Redox titrations** ($KMnO_4$, $K_2Cr_2O_7$).
11th • Org	8. Organic Chem: Basic Principles (GOC)	🔥 High (2–3 Qs)	IUPAC, **Aromaticity, Resonance & Hyperconjugation**, Acidic/Basic strength order, **Kjeldahl/Dumas estimation**.
11th • Org	9. Hydrocarbons	🔥 High (2 Qs)	Markovnikov/Anti-Markovnikov addition, **Ozonolysis**, Alkyne acidity, **Benzene EAS reactions**.
12th • Phys	1. Solutions	🔥 High (1–2 Qs)	Raoult's Law (Ideal/Non-ideal), **van't Hoff factor (i) & Colligative properties** (ΔT_b, ΔT_f, Osmotic pressure).
12th • Phys	2. Electrochemistry	🔥 High (2 Qs)	**Nernst Equation & $\Delta G°$**, **Kohlrausch's Law & Molar Conductivity**, Faraday's electrolysis laws.
12th • Phys	3. Chemical Kinetics	🔥 High (2 Qs)	**Zero & First Order integrated laws + Half-life ($t_{1/2}$)**, **Arrhenius Equation** ($k = A \cdot e^{-Ea/RT}$).
12th • Inorg	4. d- and f-Block Elements	🔥 High (2 Qs)	Lanthanoid contraction, Spin-only magnetic moment, **$KMnO_4$ & $K_2Cr_2O_7$ preparation & oxidizing reactions**.
12th • Inorg	5. Coordination Compounds	🔥 High (2–3 Qs)	**VBT & CFT Splitting (Δ_o, pairing, color, μ)**, **Structural & Stereo Isomerism**, Metal carbonyl synergic bond.
11/12 • Inorg	6. p-Block Elements (Gr 13–18 JEE)	🔥 High (1–2 Qs)	**Inert pair effect**, Anomalous behavior of B/C/N/O/F, Hydride boiling points & Oxoacid structures.
12th • Org	7. Haloalkanes & Haloarenes	🔥 High (1–2 Qs)	**S_N1 vs S_N2 reactivity & stereochemistry**, E1/E2 Zaitsev rule, Grignard reagent reactions.
12th • Org	8. Alcohols, Phenols & Ethers	🔥 High (1–2 Qs)	Lucas test, **Reimer-Tiemann & Kolbe reactions**, Williamson synthesis & **HI cleavage of Ethers**.
12th • Org	9. Aldehydes, Ketones & Carboxylic	🔥 High (2–3 Qs)	**Aldol & Cannizzaro reactions**, Tollens/Fehling/Iodoform tests, Clemmensen/Wolff-Kishner, HVZ.
12th • Org	10. Amines	🔥 High (1–2 Qs)	Basicity order in gas vs aqueous phase, **Hoffmann Bromamide, Carbylamine, Hinsberg & Diazonium salts**.

8. MATHEMATICS: Class 11 & 12 NCERT Master Matrix & Highlighted Topics

Subject 3: Mathematics

JEE Mathematics mein concept ke saath **Speed & Question-Solving Approach** sabse zaroori hai. Apni website ke Maths notes mein har chapter ke andar "**Standard JEE Question Models**" aur "**Time-Saving Shortcut Formulas**" ko highlight karein:

Class	NCERT Chapter Name	Repeat Tag & Avg Qs	Must-Highlight Topics in Website Notes (High-Frequency Concepts)
11th	1. Sets, Relations & Functions	🔥 High (1–2 Qs)	**Equivalence relations count**, Domain & Range, One-One/Onto, GIF [x] & Fractional part {x} properties.
11th	2. Trigonometric Functions	📘 Base (0–1 Q)	Allied angles, Sum-to-product identities, **Max/Min of a·sin x + b·cos x**, Conditional identities.
11th	3. Complex Numbers & Quadratics	🔥 High (2 Qs)	**Cube roots of unity (1, ω, ω²)**, Locus in Argand plane ($\|z - z_1\|$), **Newton's formula & Location of roots**.
11th	4. Permutations & Combinations	🔥 High (1–2 Qs)	**Gap & Tie method**, Dictionary rank shortcut, Distribution of identical objects (Beggar's method), Derangement.
11th	5. Binomial Theorem	🔥 High (1–2 Qs)	**General term (T_{r+1}) & Term independent of x**, Remainder problems using Binomial, Multinomial coefficients.
11th	6. Sequences & Series	🔥 High (2 Qs)	AP & GP properties, **AM ≥ GM inequality**, AGP sum, **Telescoping series (V_n method)**.
11th	7. Straight Lines	⭐ Med (1 Q)	**Image & Foot of perpendicular**, Orthocentre/Circumcentre/Centroid, Family of lines ($L_1 + \lambda L_2 = 0$).
11th	8. Conic Sections (Circle, P, E, H)	🔥 High (2–3 Qs)	Circle chords, **Focal distance, Eccentricity (e), Latus rectum & Parametric coordinates** of Parabola, Ellipse, Hyperbola.
11th	9. Limits & Derivatives (Intro)	🔥 High (1–2 Qs)	**1^∞ & 0/0 indeterminate forms**, L'Hôpital's Rule, Taylor/Maclaurin expansion series for sin x, cos x, e^x, ln(1+x).
11th	10. Statistics	🔥 High (1 Q Sure)	**Variance & Standard Deviation (σ^2)**, Effect of adding/multiplying constant on Mean & Variance, Wrong observation fix.
12th	1. Inverse Trig Functions (ITF)	⭐ Med (1 Q)	**Principal value branches**, Graphs of $\sin(\sin^{-1}x)$ vs $\sin^{-1}(\sin x)$, Telescoping $\tan^{-1}$ series.
12th	2. Matrices & Determinants	🔥 High (2–3 Qs)	**Adjoint & Inverse properties ($\|adj\ A\| = \|A\|^{n-1}$)**, Cayley-Hamilton theorem, **Cramer's Rule system of equations**.
12th	3. Continuity & Differentiability	🔥 High (1–2 Qs)	Continuity at a point, **Sharp corners & non-differentiable points** of |f(x)| and [x], Parametric/Implicit derivatives.
12th	4. Application of Derivatives (AOD)	🔥 High (2 Qs)	**Monotonicity (Increasing/Decreasing)**, **Maxima & Minima optimization**, Rolle's & Lagrange's MVT.
12th	5. Indefinite & Definite Integrals	🔥 High (2–3 Qs)	Integration by parts ($e^x[f(x)+f'(x)]$), **King's Rule $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$**, **Leibniz Rule**.
12th	6. Application of Integrals (Area)	🔥 High (1–2 Qs)	**Area bounded by Parabola & Line** ($8a^2/3m^3$ shortcut), Area with modulus & inequalities.
12th	7. Differential Equations	🔥 High (1–2 Qs)	Variable separable, Homogeneous (y = vx), **Linear Differential Equation (Integrating Factor $e^{\int Pdx}$)**.
12th	8. Vector Algebra	🔥 High (2 Qs)	Dot & Cross product geometry, Projection, **Scalar Triple Product [a b c] & Vector Triple Product a × (b × c)**.

9. Ready-to-Use Chapter Note Template (Copy-Paste Structure)

Content Workflow

Jab aap apni website mein kisi bhi chapter ke notes likhein (chahe Markdown/MDX mein ya WordPress editor mein), toh har chapter ke liye yeh exact template follow karein taaki poori website uniform aur professional dikhe:

```
--- title: "Chapter 6: System of Particles and Rotational Motion" subject: "Physics" ncert_class: 11 repeat_priority: "HIGH_REPEAT" # Displays 🔥 Most Repeated Badge (2-3 Qs/Shift) avg_questions: "2 to 3 Questions per JEE Main Shift" highlighted_topics: ["Moment of Inertia Theorems", "Angular Momentum Conservation", "Pure Rolling Motion"] --- ## 📊 1. Chapter Weightage & 5-Year JEE Trend - **Priority Level:** 🔥 High-Repeat Chapter (Must Do) - **Most Asked Sub-Topics:** 75% questions come from Moment of Inertia (with cavities) and Pure Rolling on inclined planes. ## ⚡ 2. Complete NCERT Concept Notes (With Highlighted Key Points) ### 2.1 Center of Mass (COM) of Continuous Bodies Standard theory text explaining discrete vs continuous mass distribution... <HighlightBox type="most-repeated" title="🔥 NTA Favorite Concept: Cavity Problems in COM & MOI"> Whenever a smaller disc/sphere of radius R/2 is removed from a larger body of radius R, treat the removed part as **Negative Mass (-m)**: X_com = (M₁X₁ - M₂X₂) / (M₁ - M₂) | I_remaining = I_total - I_removed </HighlightBox> <HighlightBox type="formula" title="📐 Master Formula Vault: Parallel & Perpendicular Axis"> 1. Parallel Axis Theorem (3D/2D): I = I_cm + M·d² 2. Perpendicular Axis Theorem (2D Planar Lamina ONLY): I_z = I_x + I_y </HighlightBox> <HighlightBox type="trap" title="⚠️ Common Student Trap Alert"> Never apply Perpendicular Axis Theorem to a 3D Sphere or Cylinder! It works strictly for flat 2D plates. </HighlightBox> ## 🎯 3. Top 5 Repeated PYQ Question Archetypes (Solved) 1. **Archetype 1:** Finding Moment of Inertia of a disc after cutting a circular hole...
```

10. 30-Day Step-by-Step Website Launch Action Plan

Execution Timeline

Timeline	Milestone Goal	Exact Action Steps (Bina Kisi Problem Ke Kaise Karein)
Week 1 (Days 1–7)	Website Setup & NCERT Master Index	Setup Next.js + Tailwind (or WordPress). Create the 3 Subject Dashboards (Physics, Chemistry, Maths) and input all Class 11 & 12 chapter titles with their 🔥 High Repeat badges from Pages 3–5 of this guide.
Week 2 (Days 8–15)	Upload Top 20 High-Repeat Chapters First	Saare 60+ chapters ek saath likhne mat baithiye! Pehle Physics, Chem aur Maths ke top 6–7 🔥 High Repeat chapters ke complete detailed notes + highlighted boxes upload karein.
Week 3 (Days 16–23)	Add Smart Features & Remaining Chapters	Add the **1-Click Revision Mode toggle** (CSS class toggle that hides normal paragraphs and shows only `.highlight-box` elements), KaTeX formula rendering, and upload Medium-priority chapters.
Week 4 (Days 24–30)	Mobile Testing, PDF Sheets & SEO Launch	Check mobile responsiveness (80% JEE students phone par padhte hain). Submit your sitemap to Google Search Console and share chapter links on Reddit (r/JEENEETards) & Telegram groups.

11. Essential Rules to Avoid Legal, Technical & Speed Problems

Final Checklist

🛡️ **Copyright & Content Safety**

- **No Direct Coaching Scans:** Kisi coaching institute ke PDF ya modules ka screenshot mat daalein (copyright strike aa sakti hai). Write clean digital notes following NCERT sequence.
- **Diagrams in SVG/WebP:** Physics ke pulley/circuit diagrams ko Canva, Excalidraw, ya GeoGebra par banakar lightweight SVG/WebP format mein lagayein.

⚡ **Speed, Mobile UX & Monetization**

- **Mobile-First Tables & Formulas:** Long chemical equations par `overflow-x: auto` lagayein taaki mobile screen ke bahar text na kate.
- **Clean Monetization:** Shuruat mein pop-up ads bilkul mat lagayein. Once traffic reaches 10k+ students/month, offer printable 1-page Formula Booklets or ethical sponsorships.